

Candidate Name

Centre Number

Candidate Number



For Performance Measurement

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

General Certificate of Education Ordinary Level

BUILDING TECHNOLOGY & DESIGN

PAPER 1 Theory, Drawing and Quantities.

4052/1

3 hours

SPECIMEN PAPER

Additional materials:

Answer booklet,

Plain paper **A4** (1 sheet),

Scientific calculator,

Drawing instruments.

A3 plain paper (1 sheet)

Laptop or desktop with AutoCAD software installed

INSTRUCTIONS TO CANDIDATES

Write your name, centre number and candidate number in the spaces provided at the top of this page and on **all** separate answer booklets and plain paper used.

At the end of the examination, fasten the plain paper and answer booklet securely to the question paper.

INFORMATION FOR CANDIDATES

The question paper comprises **three** sections **A**, **B** and **C**.
Marks are given in brackets [] at the end of each question.
All dimensions are in **millimetres** unless otherwise stated.

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Section A	
Section B	
Section C	
TOTAL	

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SECTION A [40 Marks]

Answer **all** questions in the spaces provided.

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- 1** **(a)** Name any **one** type of shelter used for human habitation before colonialism in Zimbabwe.
- _____ [1]
- (b)** Give any **two** advantages of using drawing software.
- _____
- _____ [2]
- (c)** Suggest any **one** source of finances for a business in the building construction industry.
- _____ [1]
- 2** **(a)** Write in full, the following abbreviations as used in the construction industry:
- (i)** **H.A.S.A.W.A.,**
- _____ [1]
- (ii)** **H.S.E.**
- _____ [1]
- (b)** Suggest any **two** ways employers promote health and safety at work places.
- _____
- _____
- _____ [2]

- 3 (a) Name the tool shown in **Figure 3(a)**.

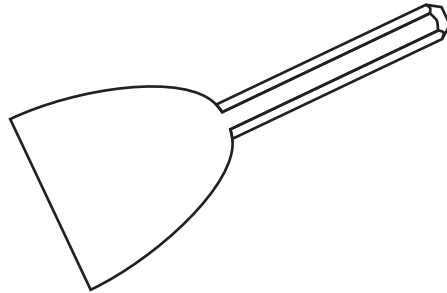


Fig. 3(a)

- (i) Tool name.
_____ [1]
- (ii) State the correct use of the tool named in (a)(i).
_____ [1]
- (b) Name any **two** tools used when setting out a circular hut.

_____ [2]

4 (a) Define the following materials used in building:

(i) fine aggregate,

_____ [1]

(ii) matrix.

_____ [1]

(b) State **two** methods of batching building materials.

_____ [2]

5 (a) Define the term *bonding* as used in bricklaying.

_____ [2]

(b) State the laps maintained in each of the following bonds:

(i) Stretcher,

_____ [1]

(ii) English.

_____ [1]

- 6 (a) List any **two** tools or equipment needed when setting out a rectangular building.

 _____ [2]

- (b) State any **two** factors considered when selecting a building site.

 _____ [2]

- 7 **Figure 7** shows a cross section of a foundation of a **one** brick wall.

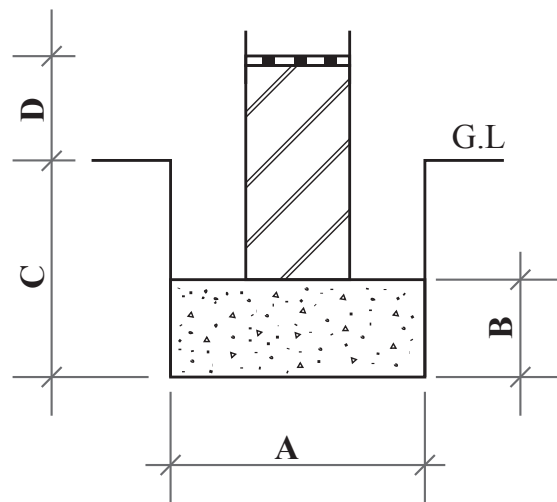


Fig. 7

State the minimum measurements represented by the letters **A** to **D**.

A _____ [1]

B _____ [1]

C _____ [1]

D _____ [1]

8 **Figure 8(a)** shows a lintel over an opening.

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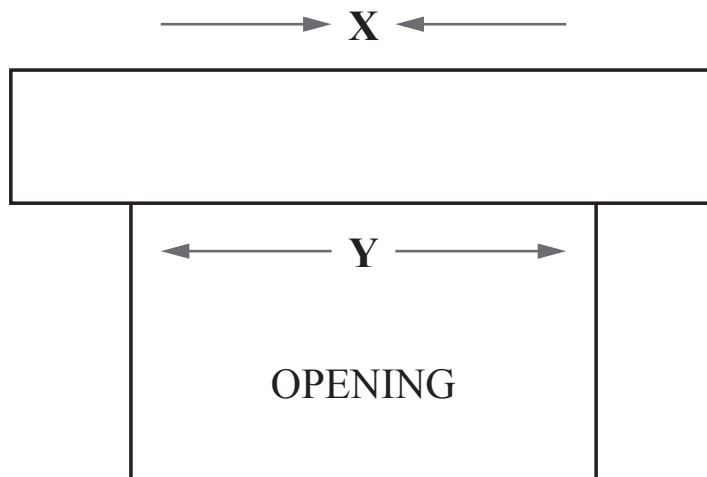


Fig. 8(a)

(a) Identify the forces **X** and **Y**.

X _____ [1]

Y _____ [1]

(b) State the functions of the following components when fixing frames to walls:

(i) lugs,

_____ [1]

(ii) props.

_____ [1]

- 9 (a) Name any **two** wall cladding materials used in the local community.

 [2]

- (b) Illustrate the following types of mortar joints:

- (i) struck,

[1]

- (ii) weather struck.

[1]

- 10 (a) Name any **two** sources of electricity.

 [2]

- (b) State any **two** safety precautions when working with electricity.

 [2]

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SECTION B [45 marks]

*Answer question 11 and any other **two** questions from this section.*

Write your answers in the separate answer booklet provided.

- 11 (a)** List any **one** duty of each of the following building construction personnel:
- (i) Quantity surveyor, [1]
 - (ii) Clerk of Works, [1]
 - (iii) Safety and Health Inspector, [1]
 - (iv) Architect, [1]
 - (v) Structural engineer. [1]
- (b)** Use any **five** points to justify the use of modern technologies in the construction of buildings. [5]
- 12 (a)**
- (i) Draw a bricklayer's trowel. [2]
 - (ii) Label any **three** parts on the diagram in **(a)(i)**. [3]
- (b)** Outline the uses of the following equipment at a building site:
- (i) site square, [1]
 - (ii) trencher. [1]
- (c)**
- (i) Draw a diagram to show the slump test. [3]
 - (ii) Label any **two** parts on the diagram in **(i)**. [2]
 - (iii) Name the **three** slump types. [3]

- 13 (a) **Figure 13** shows the plan of a **two** roomed house. Copy the diagram and show positions of profiles and ranging lines for trench width.

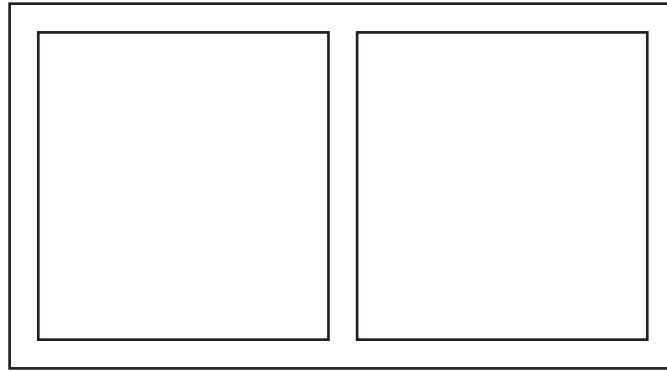


Fig. 13

[10]

- (b) Distinguish between **monolithic** and **separate** floor construction. [2]
- (c) Explain the functions of hardcore in solid ground floors. [3]
- 14 (a) **Figure 14** shows the plan outline of a straight wall in English Bond.

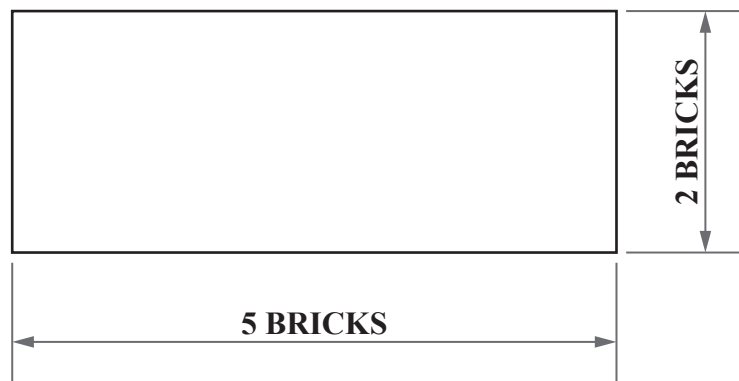


Fig. 14

Draw **two** alternate plan courses to show the correct bonding. [10]

- (b) List any **two** advantages of using thatch as a roof covering material. [2]
 - (c) Explain any **three** functions of external walls. [3]
- 15
 - (a)
 - (i) Draw a cross section of a blair toilet. [6]
 - (ii) Label any **three** parts on the diagram in (a)(i). [3]
 - (b) Outline, in sequence, the stages of fixing floor tiles. [6]

11
SECTION C [20 marks]

Answer this question.

- 16 (a) (i)** Design and draw using freehand a traditional hut of a shape of your choice. Include a window opening and a veranda. Add important dimensions to the freehand sketch. [2]

- (ii)** Draw in **First Angle** Orthographic projection, either manually or using AutoCAD:

the floor plan of the hut in **(a)(i)**,
front elevation and side elevation of the hut in **(a)(i)**. [8]

- (b)** **Figure 16(b)** shows the plan of a circular water tank made of bricks.

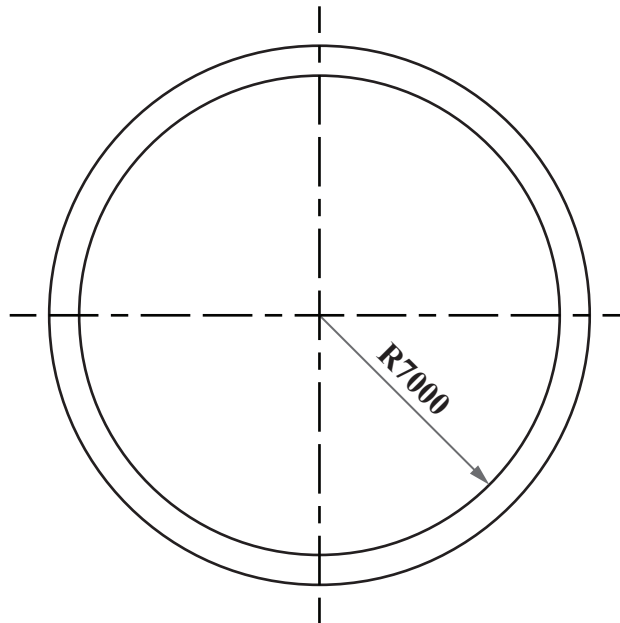


Fig. 16(b)

Given that:

Wall thickness is **230 mm**,

Height of water tank is **4 000 mm**,

104 bricks cover **one** square metre,

π is $\frac{22}{7}$.

12

Calculate:

- (i) the volume of water when the tank is **half full**, [4]
- (ii) the number of bricks used to build the super structure. [6]